IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- ## Summary of Methodologies:

This project follows these steps:

- Data Collection
- Data Wrangling
- Exploratory Data Analysis
- Interactive Visual Analytics
- Predictive Analysis (Classification)

- ## Summary of Results:

This project produced the following outputs and visualizations:

1. Exploratory Data Analysis (EDA) results
2. Geospatial analytics
3. Interactive dashboard
4. Predictive analysis of classification models

Introduction

- Project background and context

1) **The Rise of Commercial Spaceflight:** The space industry has transitioned from government-led missions to a highly competitive commercial landscape, with SpaceX at the forefront of innovation.

2) **The Falcon 9 Revolution:** As the first orbital-class reusable rocket, Falcon 9 has significantly reduced the cost of access to space.

3) **Data-Driven Insights:** Analyzing launch historical data allows us to understand the patterns behind successful missions, payload trends, and the evolution of booster landing technologies.

- Problems you want to find answers

•**Launch Success Factors:** What are the key variables (Launch Site, Orbit, Payload Mass) that influence a successful mission outcome?

•**Technological Evolution:** How has the transition between different Falcon 9 booster versions improved landing success rates over time?

•**Customer & Market Trends:** Who are the primary customers driving the demand for SpaceX launches, and what are the most common orbital requirements?

•**Cost Efficiency:** Can we identify trends in payload mass that correlate with the reusability of the boosters?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
- SpaceX API • Web scrap Falcon 9 and Falcon Heavy launch records from Wikipedia
  - Perform data wrangling - Determined labels for training the supervised models by converting mission outcomes in to training labels (0-unsuccessful, 1-successful)
  - Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

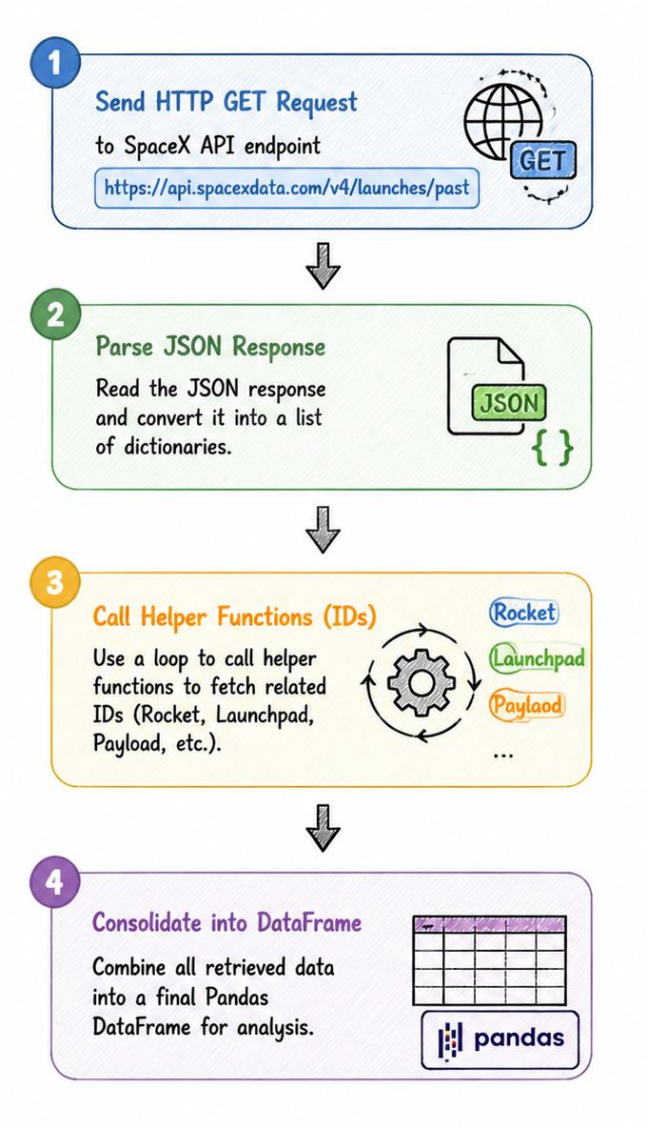
Built classification models on preprocessed data, tuned them using cross-validation and hyperparameter optimization, and evaluated their performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrix to select the best predictive model

Data Collection

- Data collection is the systematic process of gathering information from diverse sources to solve a specific research problem. For this project, data was gathered from two primary sources to predict SpaceX Falcon 9 landing success:
- SpaceX REST API: To retrieve historical launch records and technical specifications.
- Wikipedia Web Scraping: To extract supplemental data using BeautifulSoup for comprehensive analysis

Data Collection – SpaceX API

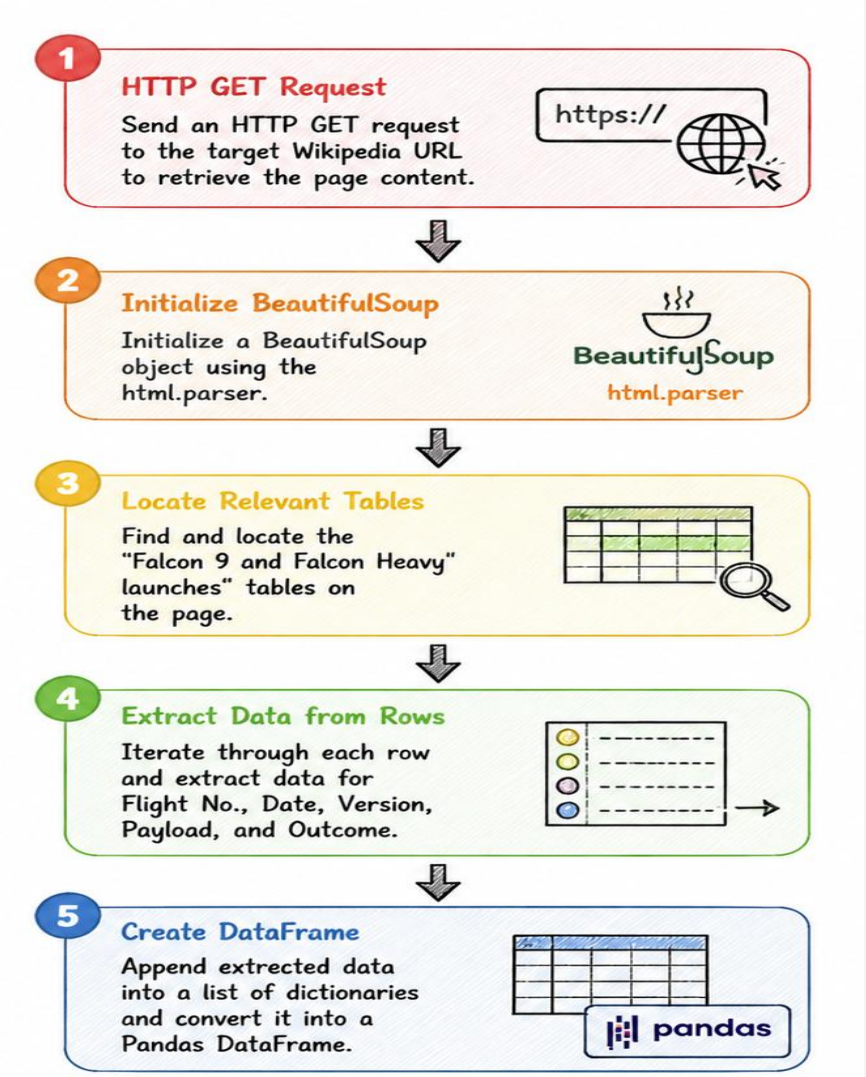
- **API Utilization:** Requested launch data from the official SpaceX REST API using the `requests` library.
- **Data Retrieval:** Fetched comprehensive data including rocket versions, launch sites, payload mass, and orbital details.
- **Feature Engineering:** Applied custom functions to convert raw JSON responses into a structured Pandas DataFrame.
- **Data Integration:** Processed complex nested API data into a flat format suitable for machine learning



View on GitHub: [SpaceX-Data-Science-Project--IBM/01-jupyter-labs-spacex-data-collection-api.ipynb at main · isuridharmasena10-gif/SpaceX-Data-Science-Project—IBM]

Data Collection - Scraping

- **HTML Extraction: Used the `requests` library to fetch the HTML content of the Wikipedia page containing Falcon 9 launch records.**
- **BeautifulSoup Parsing: Employed the `BeautifulSoup` library to navigate the HTML DOM tree and locate specific data tables.**
- **Data Parsing: Developed a robust parsing logic to extract text from HTML table cells while handling various tag structures (links, spans, etc.).**
- **Structured Storage: Organized the scraped unstructured data into a clean Pandas DataFrame for further preprocessing.**



View on GitHub: [SpaceX-Data-Science-Project--IBM/02_jupyter-labs-webscraping.ipynb at main · isuridharmasena10-gif/SpaceX-Data-Science-Project—IBM]

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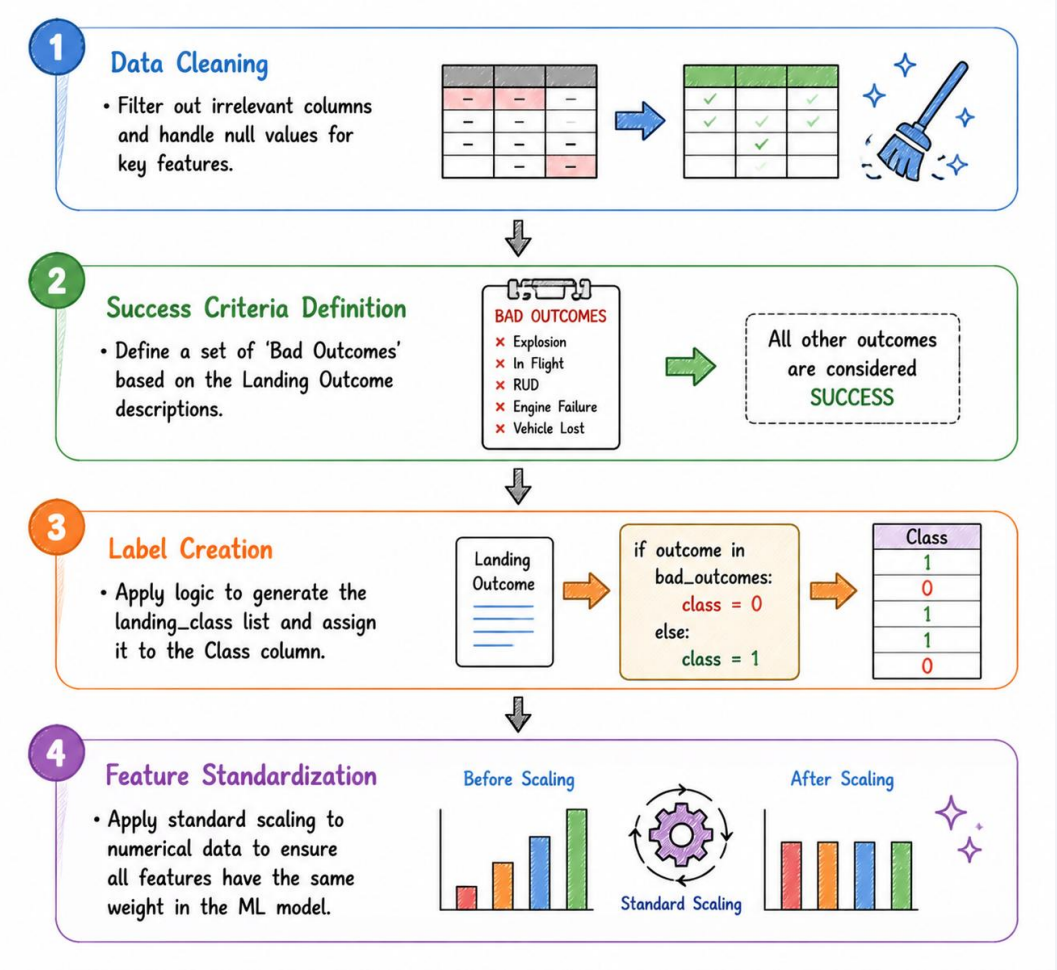
Data Wrangling

- Handling Missing Values: Identified null values in the 'PayloadMass' column and replaced them with the calculated mean to maintain data integrity.
- Outcome Categorization: Analyzed the 'Outcome' column to distinguish between successful (e.g., True ASDS) and unsuccessful (e.g., False Ocean) landings.
- Binary Label Engineering: Created a new 'Class' column where successful landings are labeled as 1 and unsuccessful landings as 0.
- One-Hot Encoding: Converted categorical features such as 'LaunchSite' and 'Orbit' into numerical dummy variables using `get_dummies`.



View on GitHub: [SpaceX-Data-Science-Project--IBM/03_spacex-Data wrangling.ipynb at main · isuridharmasena10-gif/SpaceX-Data-Science-Project—IBM]

EDA with Data Visualization

- **Objective:** To conduct a comprehensive visual inspection of the dataset using **Matplotlib** and **Seaborn** to determine which features (Launch Site, Orbit, Payload, etc.) have the strongest correlation with landing success.

List of Visualizations & Purpose:

✓ **Scatter Plot: Flight Number vs. Launch Site** - To observe the distribution of launches across different sites and identify if landing success improved as the number of flights progressed (temporal trend).

✓ **Scatter Plot: Payload Mass vs. Launch Site** - To analyze if certain launch sites are preferred for heavier payloads and to see if payload weight impacts the landing outcome at specific locations.

✓ **Bar Chart: Success Rate by Orbit Type** - To calculate and compare the mean success rate for each orbit (LEO, ISS, GTO, SSO, etc.) to determine which mission profiles are the most reliable.

✓ **Scatter Plot: Flight Number vs. Orbit Type** - To see if SpaceX's ability to land successfully in specific orbits changed over time as they gained more flight experience.

✓ **Scatter Plot: Payload Mass vs. Orbit Type** - To identify the relationship between the weight of the satellite and the target orbit, and how this combination affects the booster's ability to land.

✓ **Line Chart: Yearly Launch Success Trend** -To visualize the year-over-year improvement in technology and operational performance since the inception of the Falcon 9 program

GitHub Repository: [SpaceX-Data-Science-Project--IBM/04_eda-sql-coursera_sqllite.ipynb at main · isuridharmasena10-gif/SpaceX-Data-Science-Project--IBM]

EDA with SQL

- **Launch Site Analysis**: Identified all unique launch sites and filtered locations starting with 'CCA'.
- **Payload Metrics**: Calculated total payload for NASA (CRS) and average mass for F9 v1.1 boosters.
- **Mission Milestones**: Located the date of the first successful ground pad landing.
- **Landing Performance**: Listed boosters with successful drone ship landings (4k–6k kg payload).
- **Operational Success**: Identified boosters carrying maximum payload using SQL subqueries.
- **Failure Tracking**: Analyzed 2015 drone ship landing failures by booster and site.
- **Outcome Ranking**: Ranked all landing types by frequency between 2010 and 2017.

GitHub Repository: [SpaceX-Data-Science-Project--IBM/05_EDA_data_visualization.ipynb at main · isuridharmasena10-gif/SpaceX-Data-Science-Project--IBM]

Build an Interactive Map with Folium

Category	Folium Objects	Technical Purpose
Base Map	`Map`, `Marker`, `Circle`	To initialize the geographical view and pinpoint exact launch site coordinates.
Clustering	`MarkerCluster`	To visualize the density of success vs. failure outcomes at each site using color-coding.
Interaction	`MousePosition`	To enable real-time coordinate tracking for calculating distances to nearby infrastructure.
Analysis	`PolyLine`, `DivIcon`	To draw measurement lines and display calculated distance labels (km) to coastlines and cities.

[SpaceX-Data-Science-Project--IBM/06_Interactive Visual Analytics with Folium.ipynb at main · isuridharmasena10-gif/SpaceX-Data-Science-Project--IBM]

Build a Dashboard with Plotly Dash

Summary of Plots & Interactions:

- **Pie Chart (Launch Site Success):** An interactive chart that updates based on the selected launch site to show the ratio of successful vs. failed landings.
- **Scatter Plot (Payload vs. Success):** A dynamic plot showing the relationship between payload mass and landing outcome, color-coded by booster version.
- **Dropdown Menu (Site Selection):** Allows users to filter the entire dashboard by specific launch sites or view aggregate data for "All Sites."
- **Range Slider (Payload Mass):** Enables real-time filtering of data based on payload weight (0kg to 10,000kg) to observe how mass affects success probability.

These interactive elements were added to provide a **self-service analytics tool** for stakeholders. Instead of static reports, users can test their own hypotheses regarding how site selection and payload weight impact mission success.

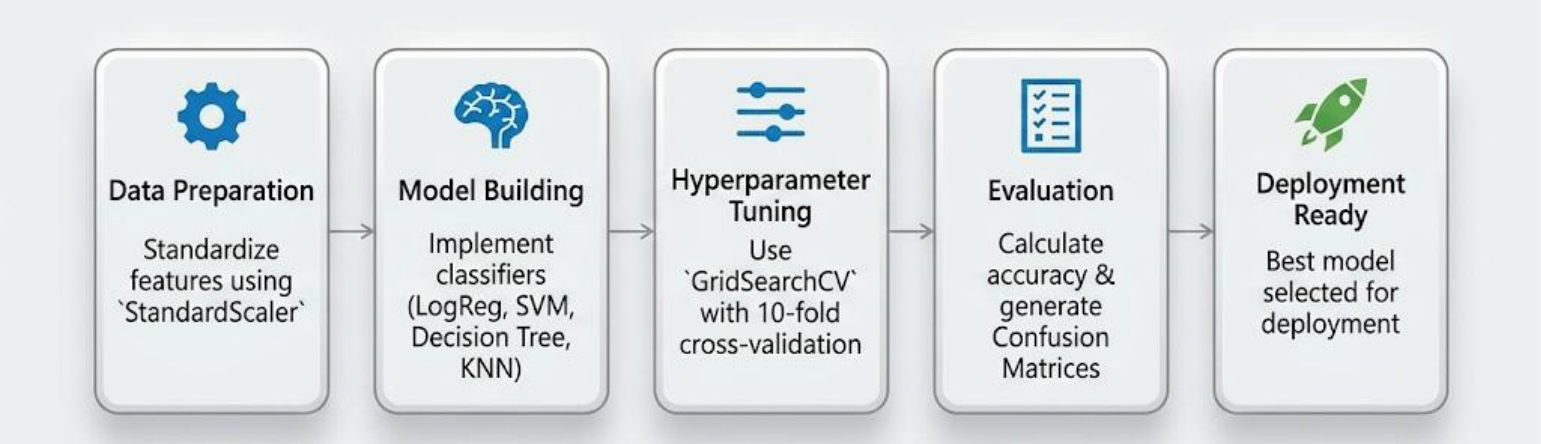
GitHub Repository: [SpaceX-Data-Science-Project--IBM/07_Interactive Dashboard with Plotly Dash.py at main · isuridharmasena10-gif/SpaceX-Data-Science-Project--IBM]

Predictive Analysis (Classification)

- **Methodology Summary:**

- **Data Preparation:** Standardized the feature data using `StandardScaler` to ensure all variables (like Payload Mass and Flight Number) were on the same scale.
- **Model Building:** Implemented four distinct classification algorithms: **Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)**.
- **Optimization:** Utilized `GridSearchCV` with 10-fold cross-validation to perform hyperparameter tuning, ensuring each model operated at its peak performance.

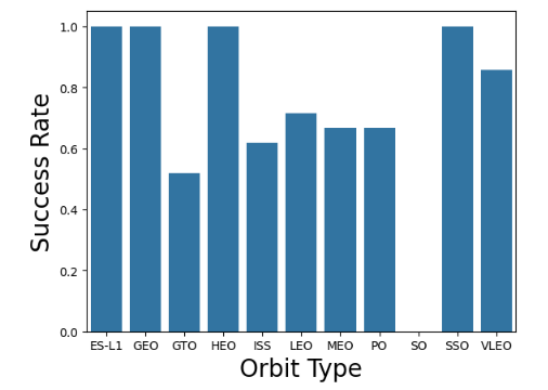
- **Evaluation:** Assessed each model using **Accuracy Scores** on the test set and generated **Confusion Matrices** to visualize true positives and false negatives.

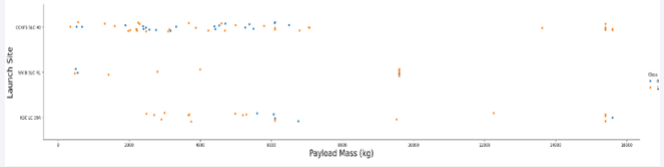


GitHub Repository

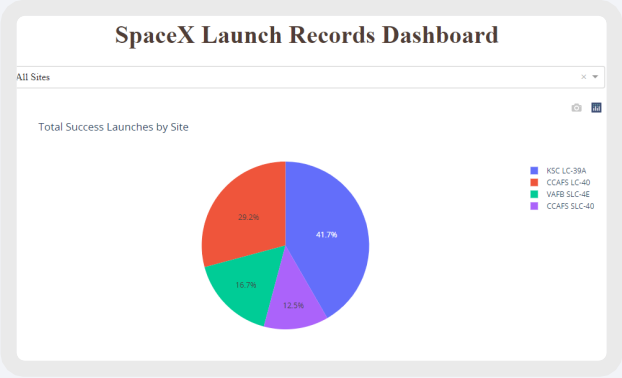
Results

- Exploratory data analysis results

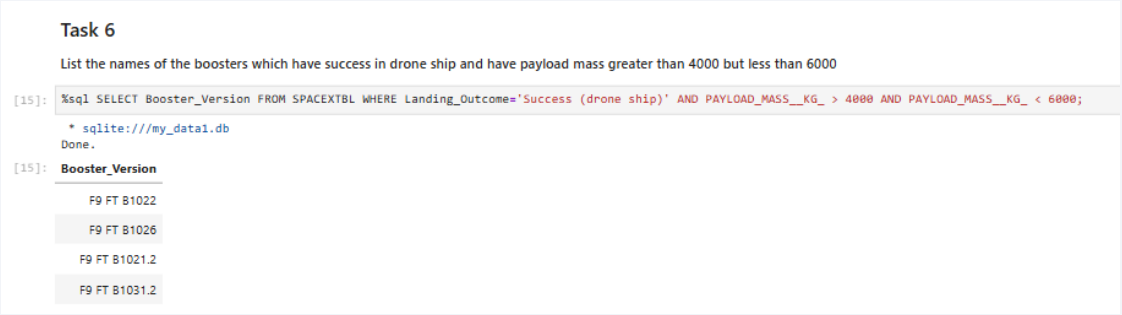




Interactive analytics demo in screenshots



Predictive analysis results



Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[15]: %sql SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome='Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
 * sqlite:///my_data1.db
Done.
```

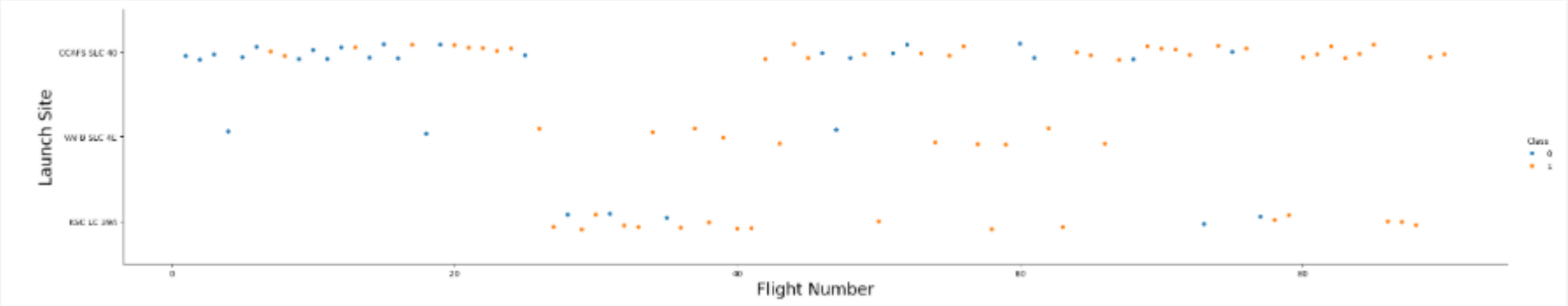
[15]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Section 2

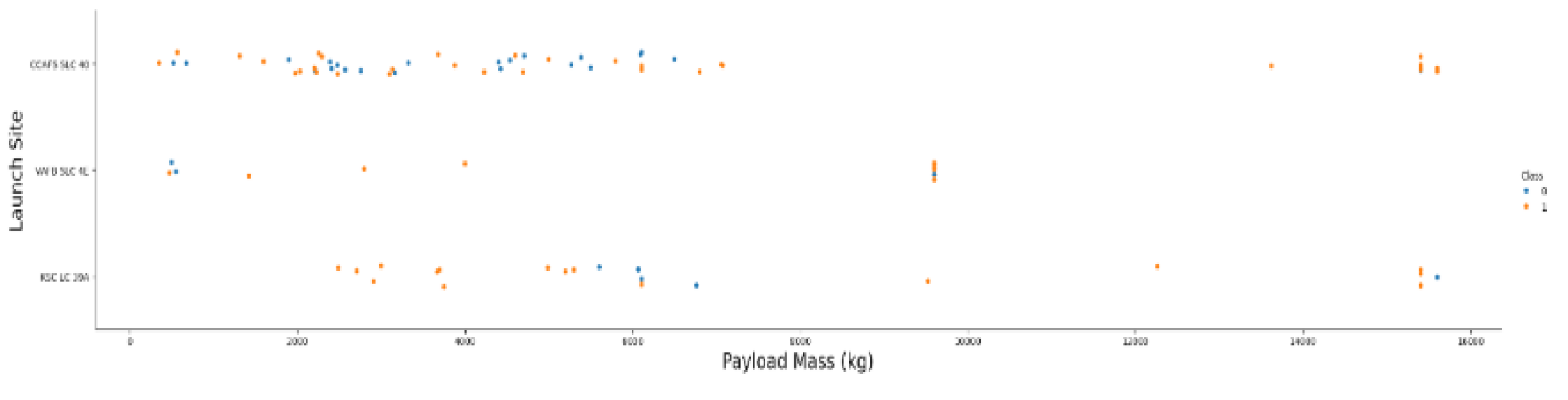
Insights drawn from EDA

Flight Number vs. Launch Site



The **Flight Number vs. Launch Site** scatter plot demonstrates the clear correlation between operational experience and landing success. While early flights across all sites showed mixed outcomes, there is a distinct transition to consistent successful landings (Class 1) as the flight numbers increase, particularly at the KSC LC 39A facility.

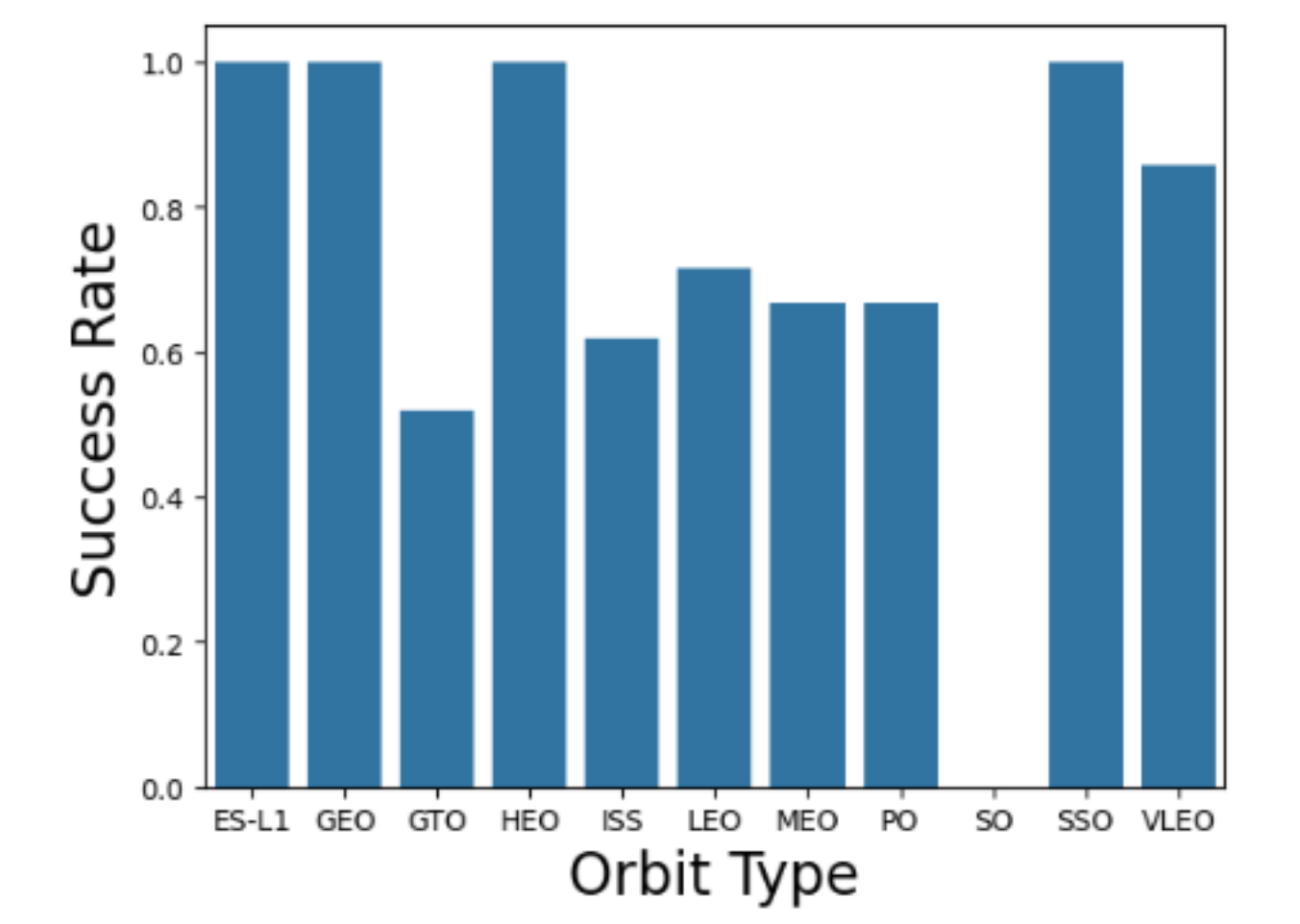
Payload vs. Launch Site



- **Mass Correlation:** The plot shows that success is not strictly limited by weight, but **KSC LC 39A** demonstrates the highest reliability for medium payloads (2,000–5,000 kg), while heavy payloads (15,000+ kg) are successfully handled at both **KSC LC 39A** and **CCAFS SLC 40**.

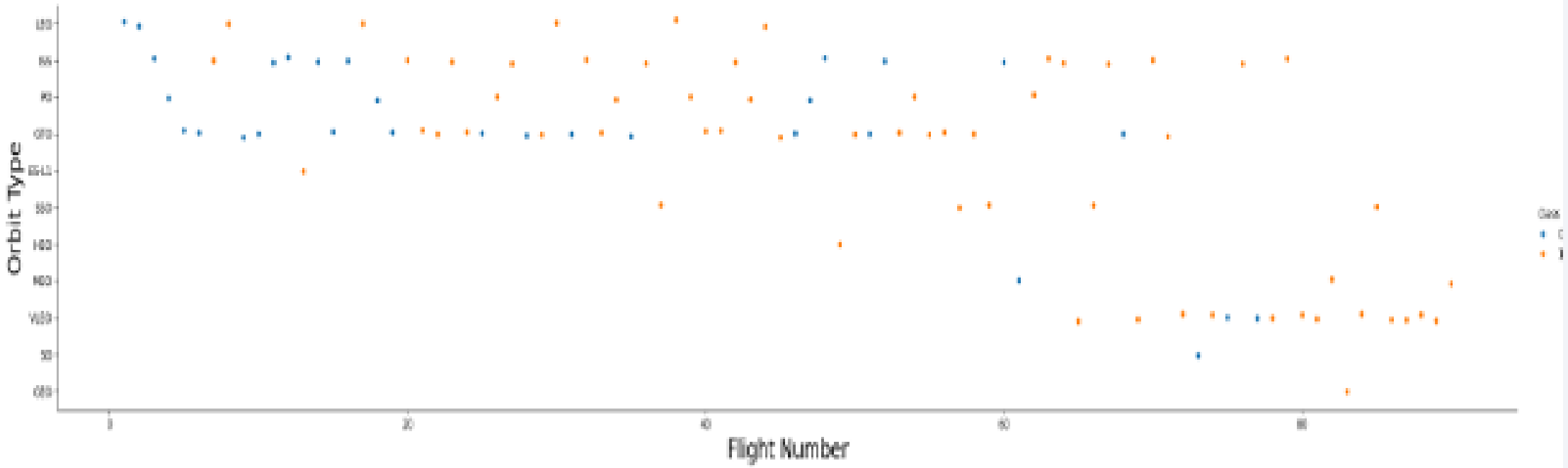
- **Site Reliability:** While **CCAFS SLC 40** handled the widest variety of payload masses with mixed early results, all sites show a transition to consistent successful landings (orange markers) as missions progressed, regardless of the payload weight

Success Rate vs. Orbit Type



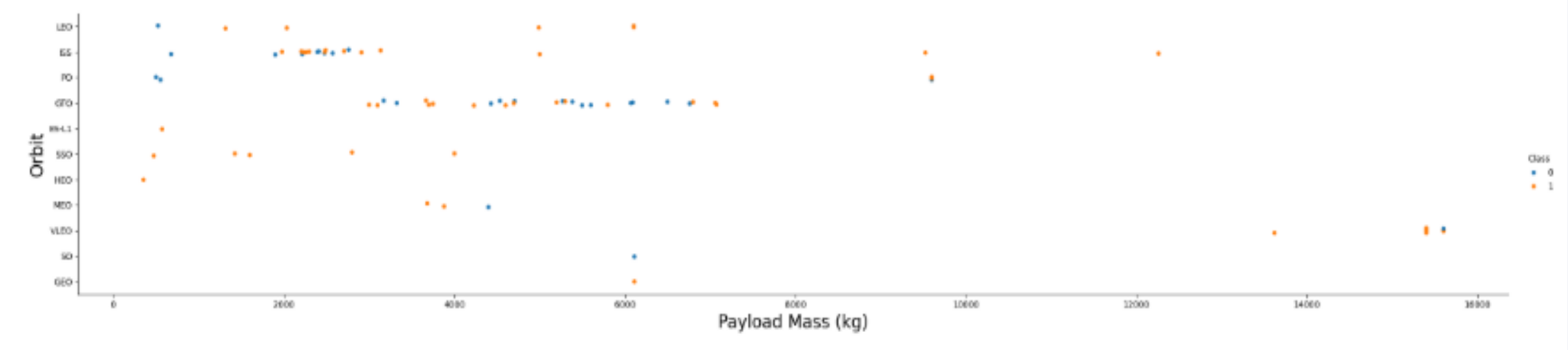
- The bar chart was plotted to identify the most reliable mission profiles, revealing that orbits like **ES-L1, GEO, HEO, and SSO** achieved a **100% success rate**, while **GTO** and **ISS** showed lower performance.

Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

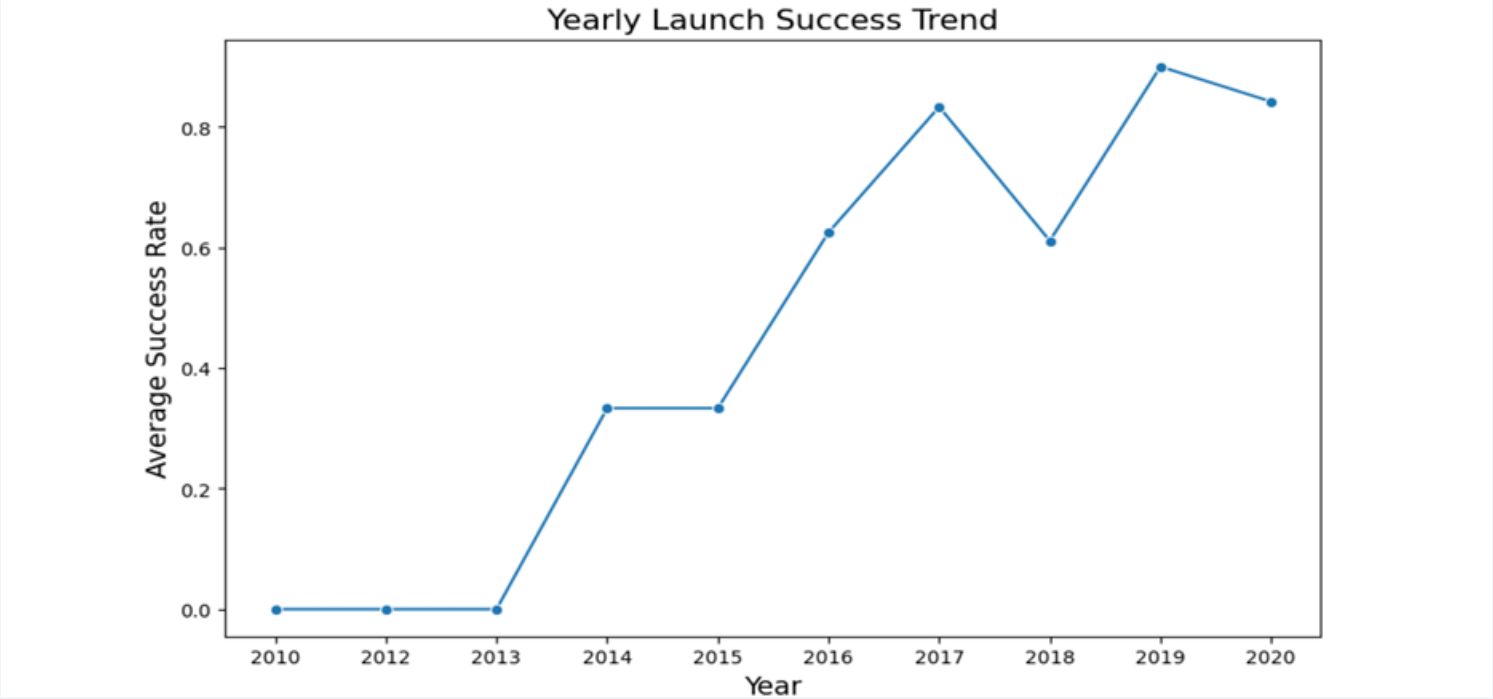
Payload Mass vs. Orbit Type



Observation: The plot illustrates that landing success (orange markers) is highly frequent across most orbits, though success becomes more scattered at lower payload masses (under 2,000 kg) in orbits like **LEO** and **PO**.

Key Insight: For heavy-payload missions (above 10,000 kg), which are predominantly directed toward **VLEO** and **GTO** orbits, SpaceX demonstrates near-perfect landing reliability, suggesting that mass is not a limiting factor for reusability in high-capacity missions.

Launch Success Yearly Trend



- The charts illustrate a significant upward trajectory in landing reliability, beginning at a **0% success rate** prior to 2013 and steadily climbing to a peak of approximately **90% by 2019**.
- The transition from flat, early-stage testing to a sustained high success rate between 2017 and 2020 confirms that the Falcon 9 has evolved into a highly mature and consistently reliable reusable launch system.

All Launch Site Names

Task 1 ¶

Display the names of the unique launch sites in the space mission

[10]:
```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTBL;
```

```
 * sqlite:///my_data1.db
Done.
```

[10]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- The unique launch site names are **CCAFS LC-40**, **VAFB SLC-4E**, and **KSC LC-39A**.

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

[11]:
```
%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

[11]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The query successfully retrieved the first five mission records from **CCAFS LC-40**, including details for early **Dragon** spacecraft qualifications and **NASA (CRS)** missions.

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

[12]:
```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer='NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

[12]:

SUM(PAYLOAD_MASS__KG_)
45596

The total payload mass carried by all boosters launched for **NASA (CRS)** missions is **45,596 kg**.

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

[13]:
```
%sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version='F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

[13]:

AVG(PAYLOAD_MASS__KG_)
2928.4

The average payload mass carried by the **F9 v1.1** booster version is **2,928.4 kg**.

First Successful Ground Landing Date

The MIN keyword is used to calculate the minimum of the DATE column, i.e. the first date, and the WHERE keyword (and the associated condition) filters the results to only the successful ground pad landings.

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

[14]:
```
%sql SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome='Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

[14]:

MIN(Date)
2015-12-22

- The first successful landing outcome on a ground pad was achieved on **2015-12-22**.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

[15]:
```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome='Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

[15]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

The SQL query result for Task 6 identifies four booster versions that successfully landed on a drone ship with a payload mass between 4,000 kg and 6,000 kg:

- **F9 FT B1022**
- **F9 FT B1026**
- **F9 FT B1021.2**
- **F9 FT B1031.2**

Total Number of Successful and Failure Mission Outcomes

The COUNT keyword is used to calculate the total number of mission outcomes, and the GROUPBY keyword is also used to group these results by the type of mission outcome.

List the total number of successful and failure mission outcomes

[16]:
```
%sql SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

[16]:

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

The dataset contains a total of **99 successful mission outcomes**, with only one recorded in-flight failure and one instance where the payload status remained unclear.

Boosters Carried Maximum Payload

The SQL query, identifies twelve Booster Versions that have carried the maximum payload mass recorded in the dataset:

```
[17]: %sql SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);

 * sqlite:///my_data1.db
Done.
```

[17]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

The WHERE keyword is used to filter the results for only failed landing outcomes, AND only for the year of 2015.

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[18]: %sql SELECT SUBSTR(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE Landing_Outcome='Failure (drone ship)' AND SU
```

```
 * sqlite:///my_data1.db
Done.
```

[18]:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

The SQL query result indicates that in 2015, two missions from launch site **CCAFS LC-40** experienced a landing failure on a drone ship: booster version **F9 v1.1 B1012** in January (01) and **F9 v1.1 B1015** in April (04).

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

[19]:
```
SELECT Landing_Outcome, COUNT(*) AS Count FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

[19]:

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

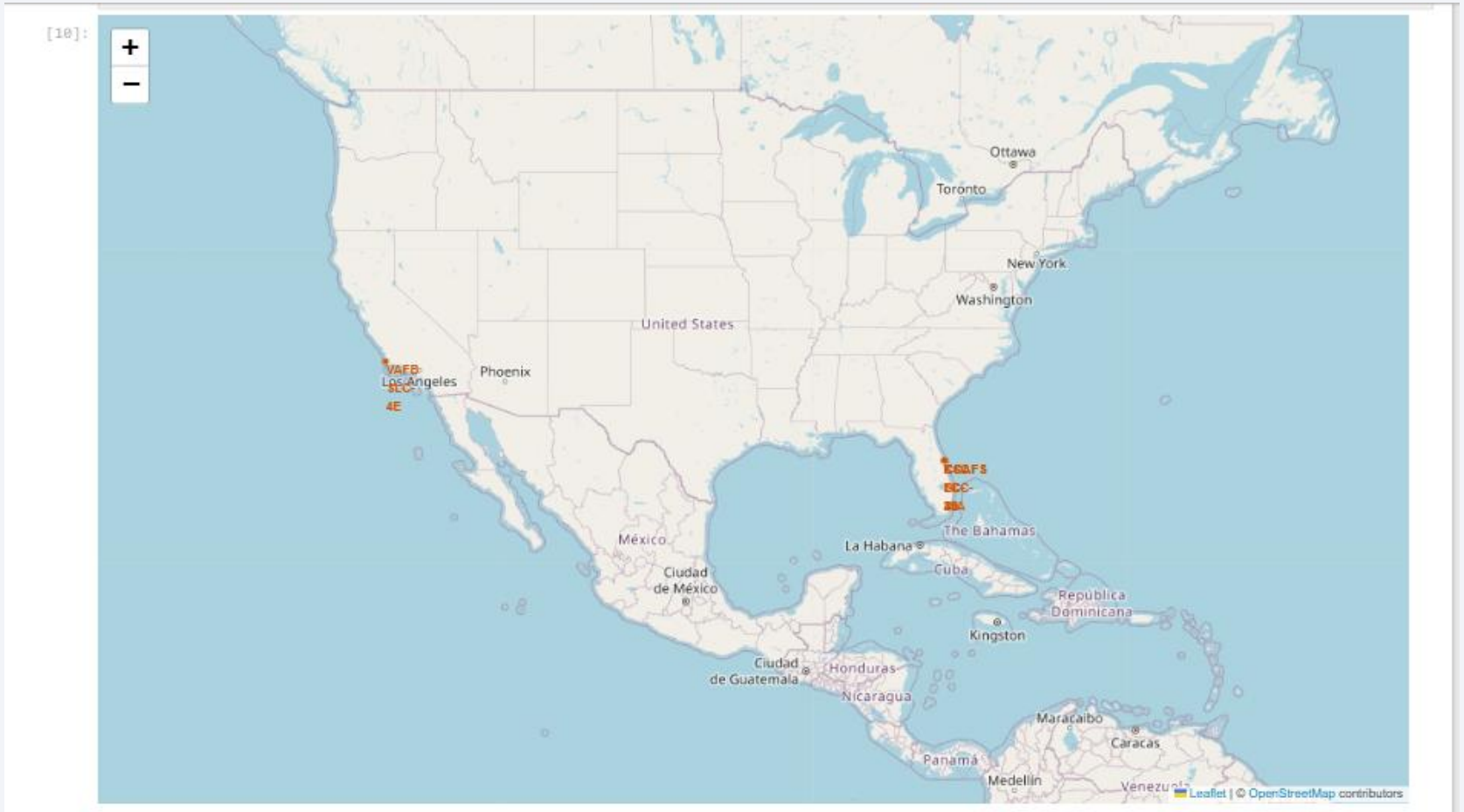
The WHERE keyword is used with the BETWEEN keyword to filter the results to dates only within those specified. The results are then grouped and ordered, using the keywords GROUP BY and ORDER BY, respectively, where DESC is used to specify the descending order.

Between **2010-06-04** and **2017-03-20**, the most frequent landing outcome was **"No attempt"** with 10 occurrences, followed by 5 successful and 5 failed drone ship landings.

Section 3

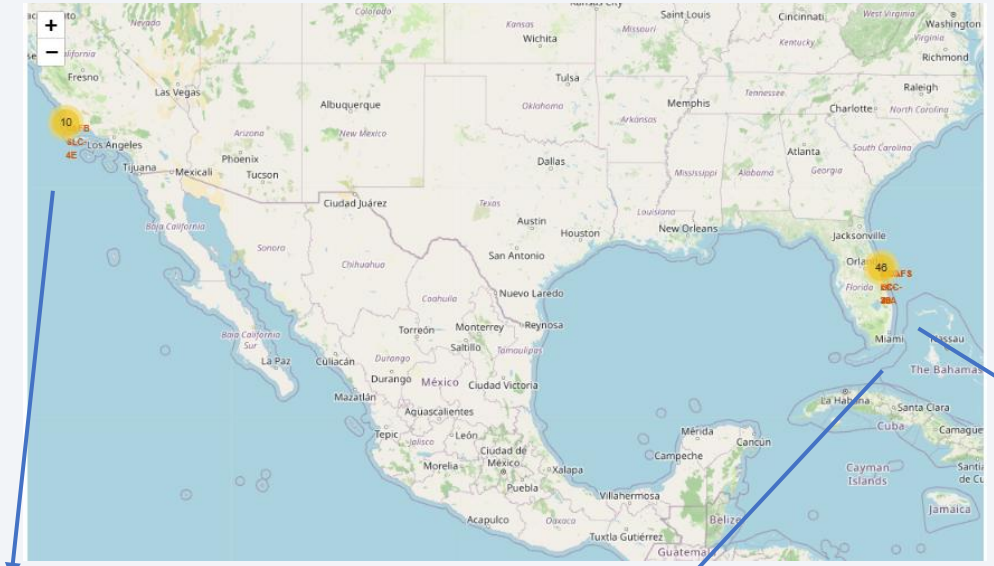
Launch Sites Proximities Analysis

All Launch Sites on a Map

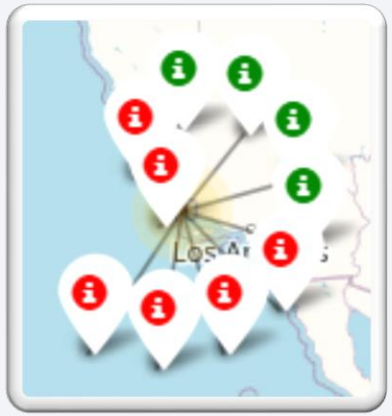


All SpaceX launch sites are on coasts of the United States of America, specifically Florida and California

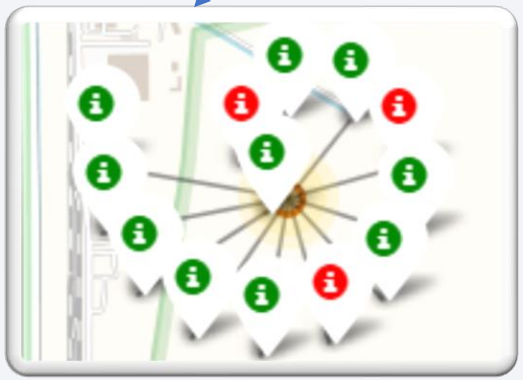
Success/failed launches for each site on the map



VAFB SLC-4E



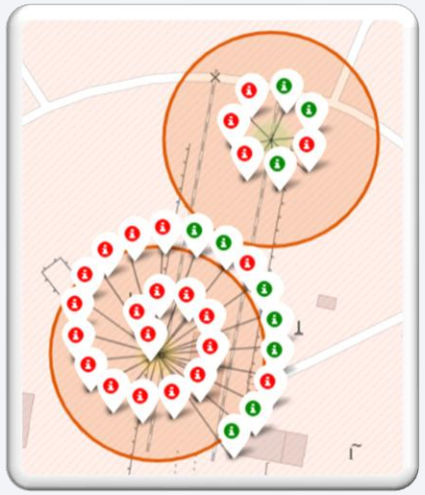
KSC LC-39A



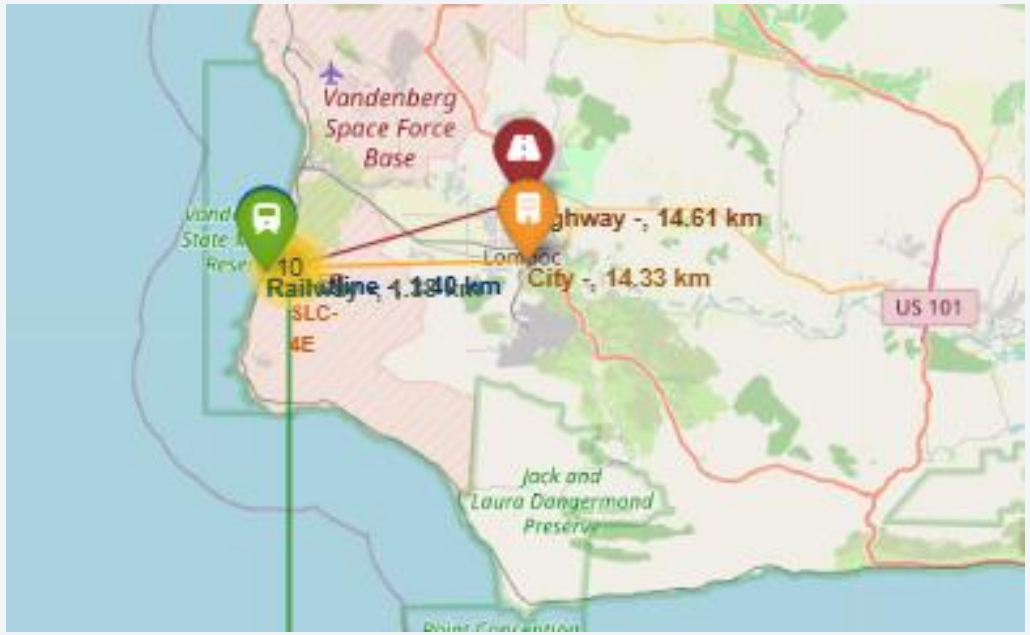
- Launches have been grouped into clusters, and annotated with green icons for successful launches, and red icons for failed launches.

CCAFS SLC-40 and CCAFS LC-40

=



The distances between a launch site to its proximities



Logistical Proximity: Facilities are situated within 0.5–2.0 km of highways, railways, and coastlines to facilitate the movement of large rocket components.

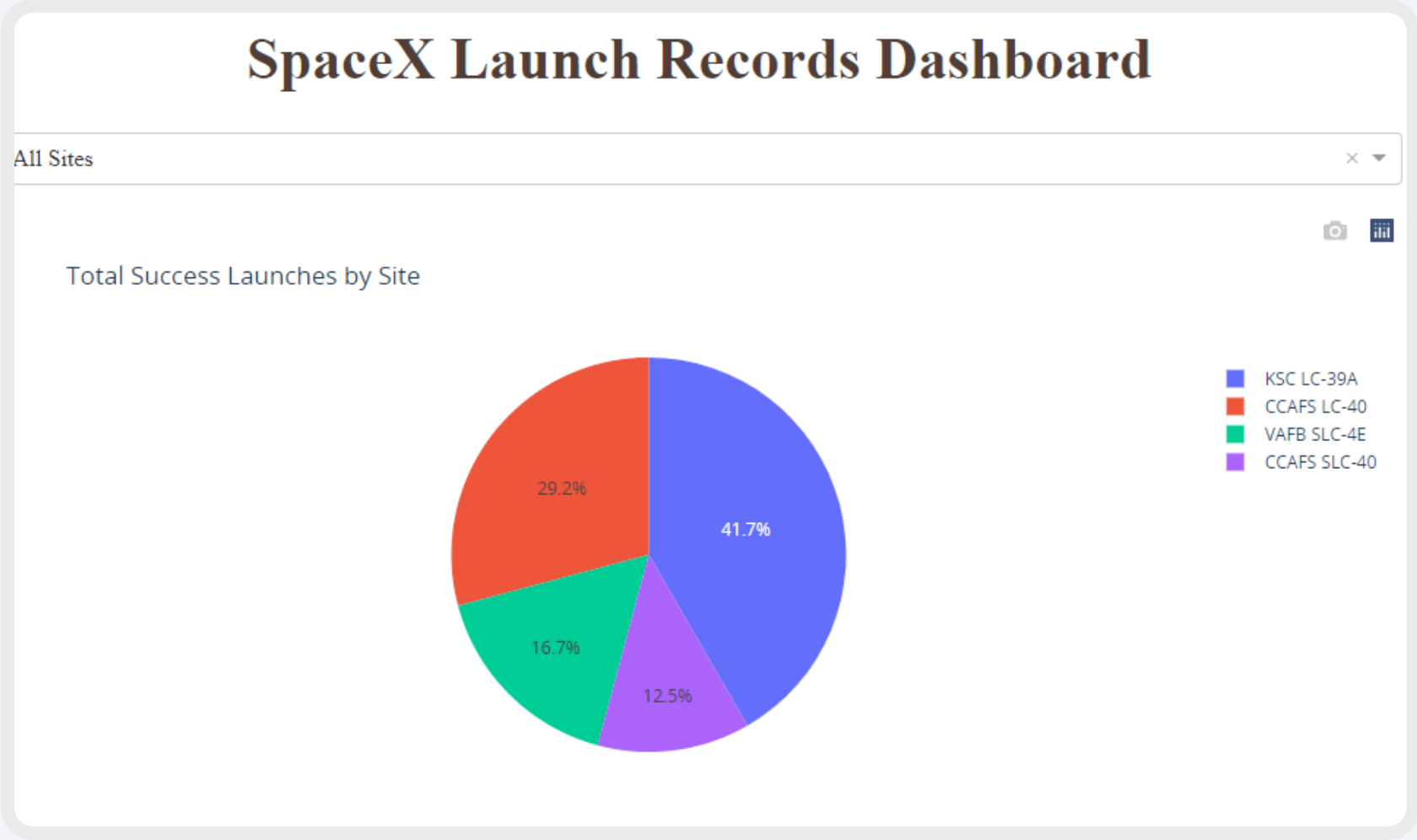
Safety Buffers: Sites are located 14 km to 72 km away from cities to mitigate risks to populated areas, ensuring flight trajectories remain primarily over the ocean

Section 4

Build a Dashboard with Plotly Dash

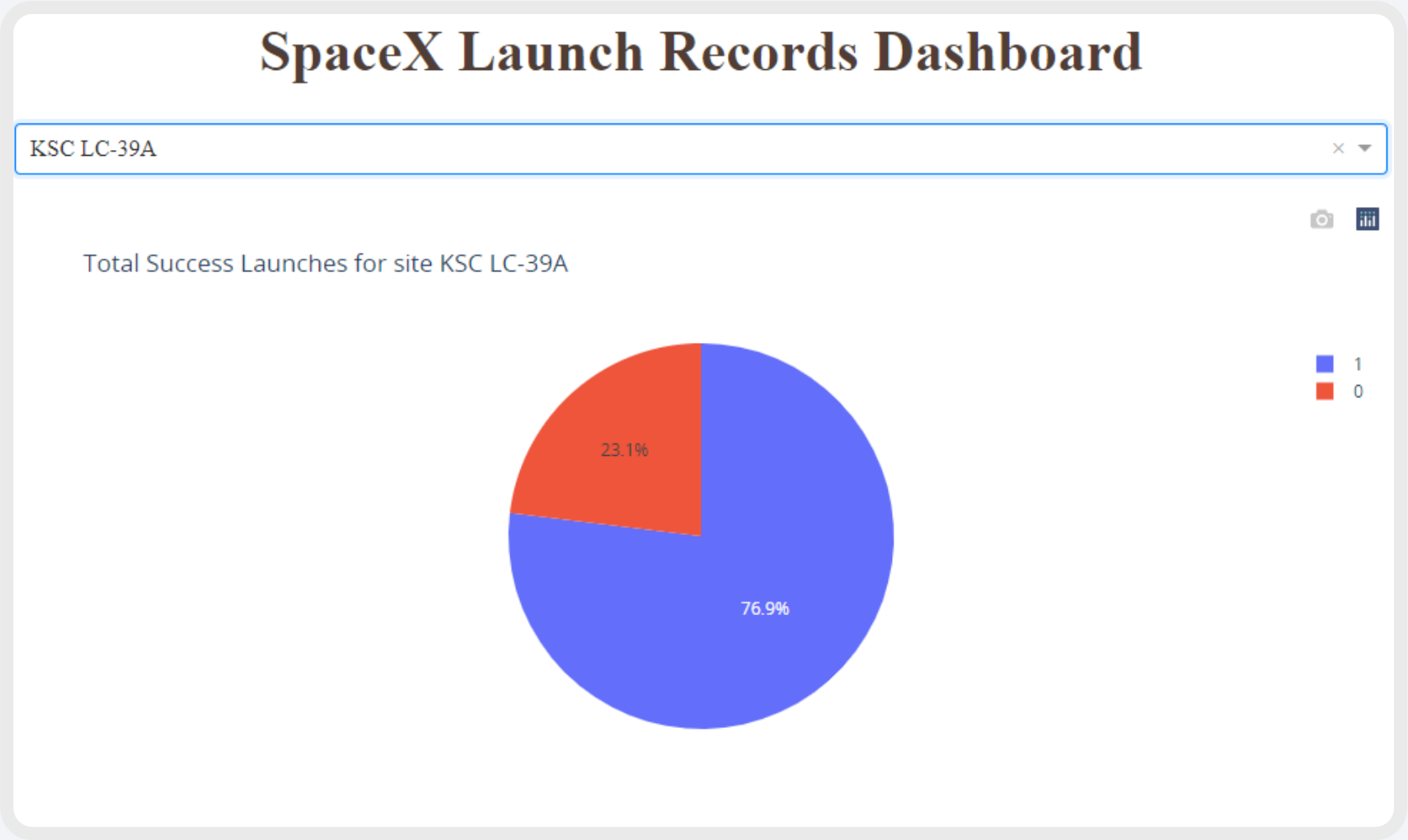
Launch success count for all sites

- The launch site KSC LC-39 A had the most successful launches, with 41.7% of the total successful launches.

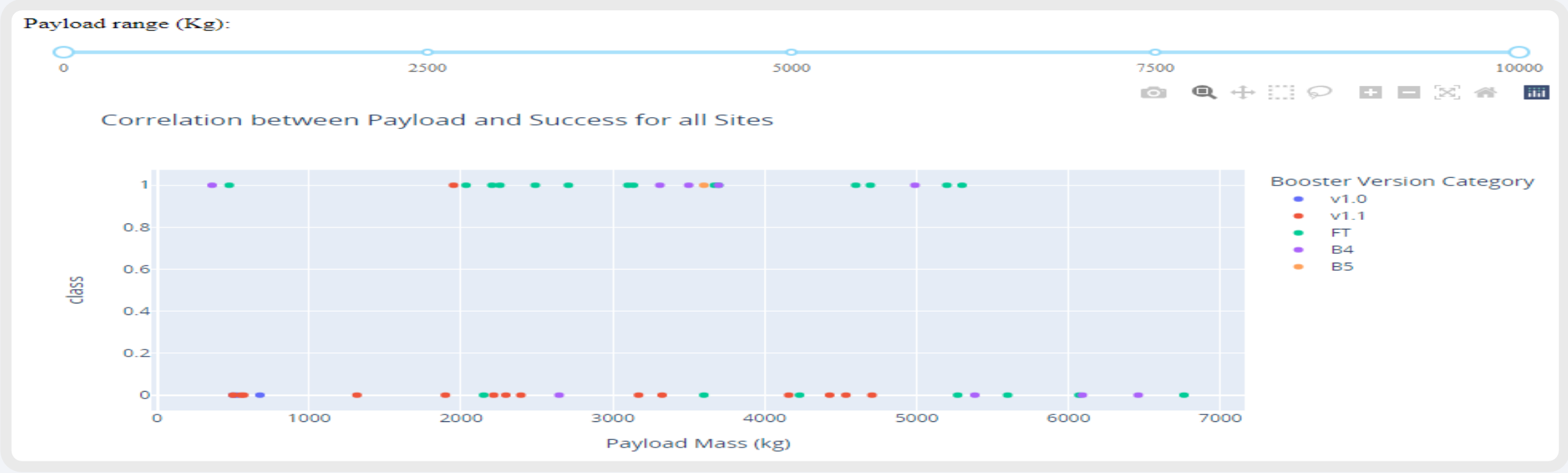


Pie chart for the launch site with highest launch success ratio

The launch site KSC LC-39 A also had the highest rate of successful launches, with a 76.9% success rate.



Correlation between payload and success

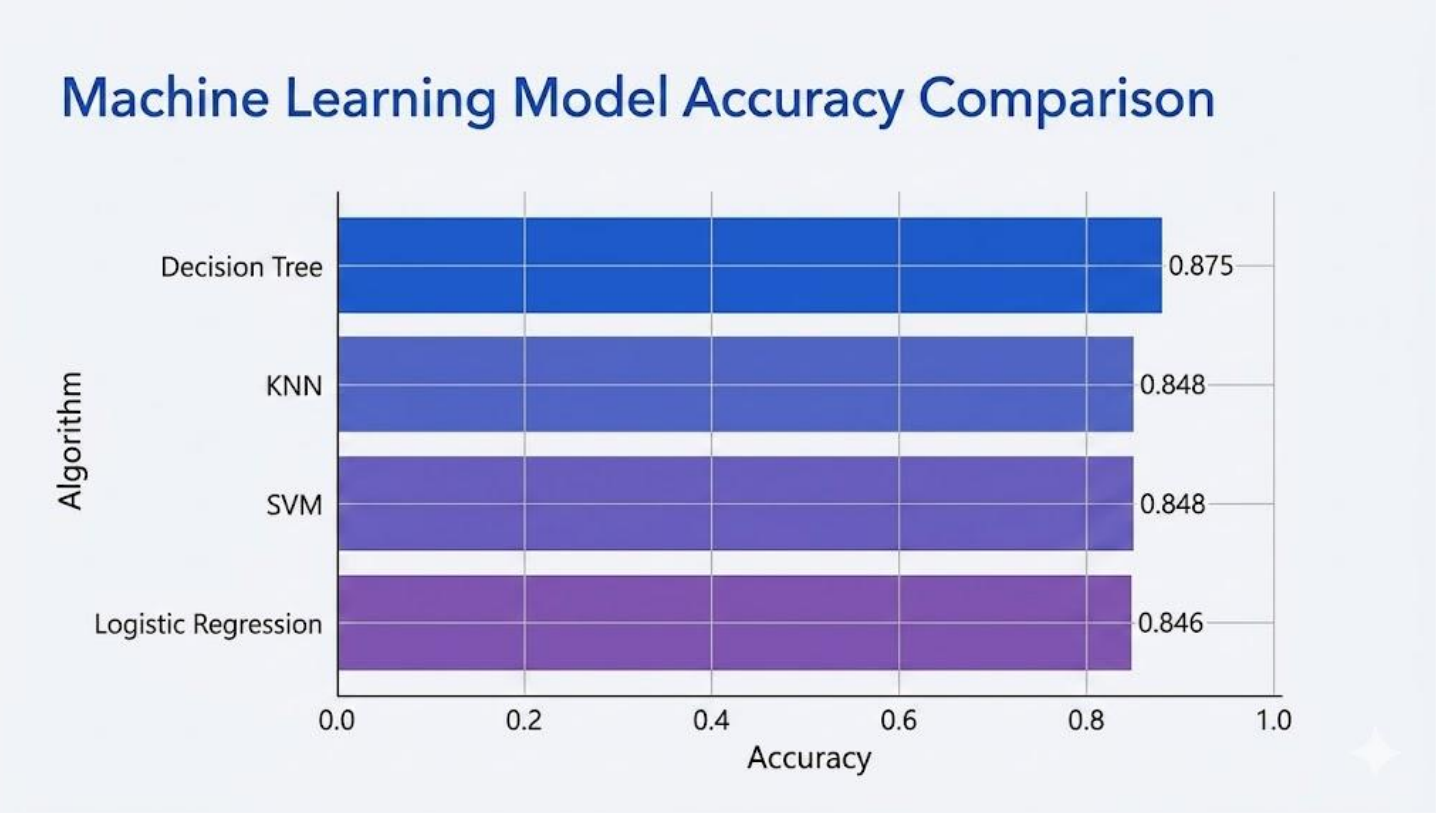


The dashboard shows no strong direct correlation between payload mass and launch success, as both successes and failures occur across different payload ranges. Booster version appears to influence success more significantly than payload weight alone.

Section 5

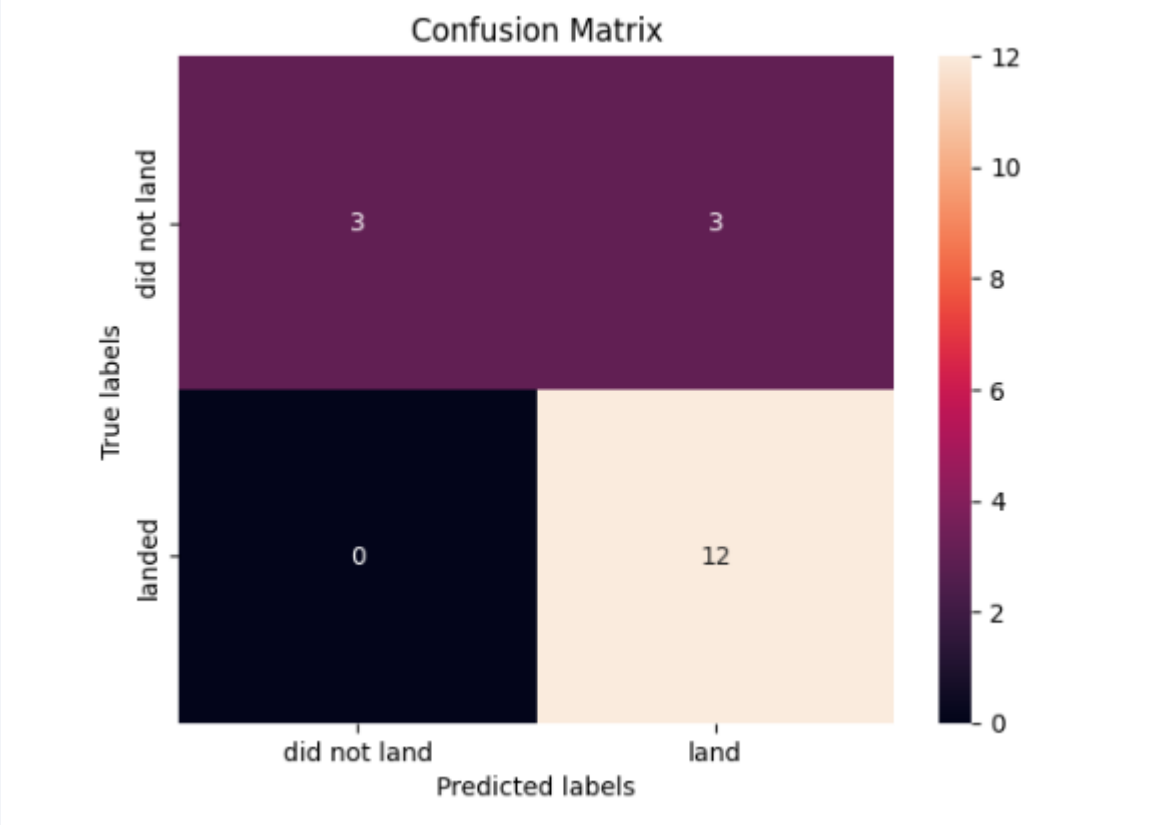
Predictive Analysis (Classification)

Classification Accuracy



- The Decision Tree model has the highest classification accuracy

Confusion Matrix



The analysis of the confusion matrices across all classification models reveals an identical performance pattern:

•**Successful Landing Predictions (True Positives):** The models correctly identified **12 instances** where the booster successfully landed.
•**Incorrect Success Predictions (False Positives):** There were **3 instances** where the models predicted a successful landing, but the booster actually failed to land.

This consistency indicates that while the models are highly effective at identifying successes, they share a slight tendency to misclassify specific failed outcomes as successful.

Conclusions

- Operational Learning: Landing success has surged over time, evolving from early experimental failures to sustained 90% peak reliability.
- Infrastructure Impact: Launch site and booster version are primary drivers; KSC LC-39A remains the most successful operational hub.
- Predictive Power: With 87.5% accuracy, classification models successfully forecast outcomes based on historical flight profiles.
- Business Value: Data-driven modeling supports mission planning, substantial risk reduction, and competitive cost optimization

Appendix

Notebooks & Code

Complete source code for Data Collection (API/Scraping), EDA, and Predictive Classification models.

SQL & Datasets

All structured SQL queries and raw/processed CSV datasets used throughout the project lifecycle.

Interactive Assets

Folium map HTML files and Plotly Dash scripts for dynamic geographic and payload analysis.

[isuridharmasena10-gif/SpaceX-Data-Science-Project--IBM]

Thank you!